

Udemy Course: <https://cognizant.udemy.com/course/the-complete-web-development-bootcamp/learn/lecture/12383788#overview>

Portfolio website: https://youtu.be/0YFrGy_mzjY

Dark Mode Toggle BTN:

- <https://youtu.be/NqSM99bxIug>
- <https://youtu.be/9LZGB30LXNQ>

HTML Boiler Plate:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Demo1 Website</title>
  <link rel="stylesheet" href="C:\Users\styles.css"> //style sheet CSS
  <link rel="stylesheet" href="../CSS/styles.css"> //Relative path

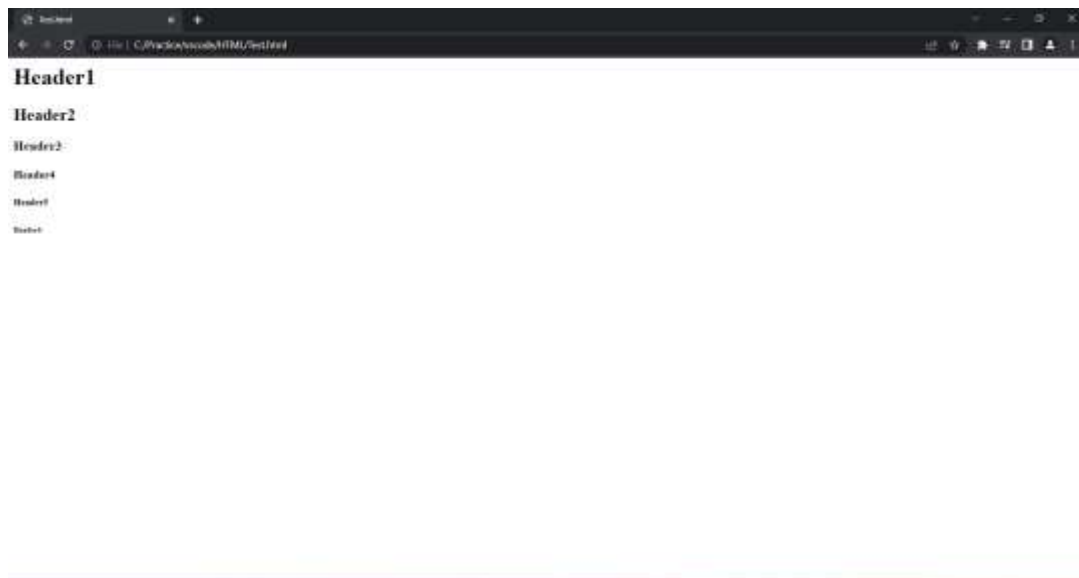
</head>
<body>
  <h1>Hello World</h1>
</body>
</html>
```

```
<!-- Comment --> // Short cut for comments : Ctrl + /
```

```
<div class="container"></div> // Short cut for container : .container
```

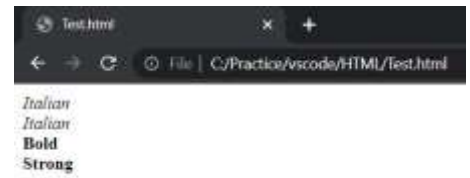
Headers:

```
<h1> Header1 </h1>
<h2> Header2 </h2>
<h3> Header3 </h3>
<h4> Header4 </h4>
<h5> Header5 </h5>
<h6> Header6 </h6>
```



Text Editing:

```
<em> Italian </em><br>
<i> Italian </i><br>
<b> Bold </b><br>
<strong> Strong </strong>
```



Anchor Tag:

```
<a href="https://www.google.com"><strong><em>Google</em></strong></a>
```

Syntax:

```
<tag attribute = value another_attribute = value> Content </tag>
```

Anchor tags in Lists:

```
<ul>
  <li><a href="https://www.google.com"><strong><em>Google</em></strong></a></li>
  <li><a href="https://www.amazon.com">Amazon</a></li>
  <li><a href="https://www.flipkart.com">Flipkart</a> </li>
</ul>
```

```
<ol type="i">
  <li><a href="https://www.google.com"><strong><em>Google</em></strong></a></li>
  <li><a href="https://www.amazon.com">Amazon</a></li>
  <li><a href="https://www.flipkart.com">Flipkart</a> </li>
</ol>
```

```
<ol start="5">
  <li><a href="https://www.google.com"><strong><em>Google</em></strong></a></li>
  <li><a href="https://www.amazon.com">Amazon</a></li>
  <li><a href="https://www.flipkart.com">Flipkart</a> </li>
</ol>
```

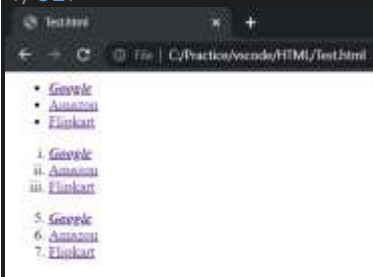


Image hovering:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Hovering</title>
    <link rel="stylesheet" href="C:\Practice\vscode\CSS\test.css">
  </head>
  
```

In css file

```
body{
  background-color: #252525;
  margin:0;
}
img:hover{
  background-color: white;
  height:100px;
}
.circular{
  border-radius: 100%;
  height:100px;
}
```

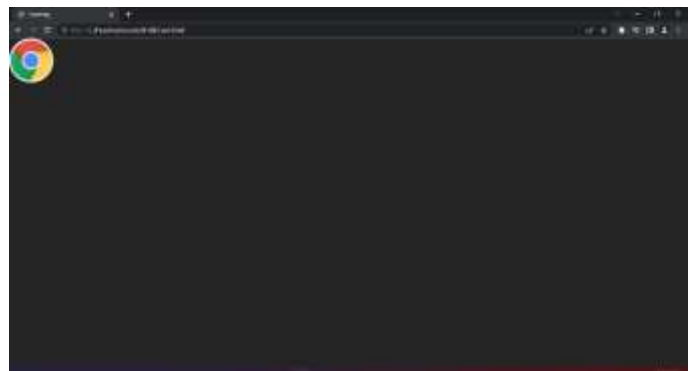
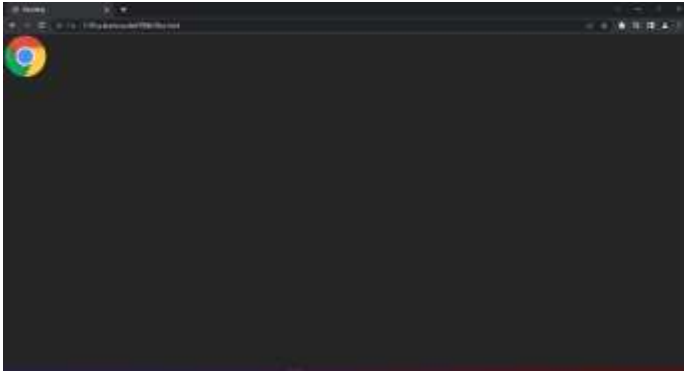


Image Attribute:

Photo bucket – Image Uploading Website

Webpages:

Image anchor tag ->

```
<a href="https://www.google.com"></a>
```

S4V29: GitHub

S5V32: CSS (Cascading Style Sheet)

Adding CSS will be 3 types

- Inline
- Internal
- External

->Inline: <tag style="css" />

```
<html style="background:black"></html>
```

Html-tag

Style-name

Background-Property

Black-Value

➔ Helps to target only a single element

->Internal: <style>css</style>

```
<html>
  <head>
    <style>
      html{background:black;
    }
  </style>
</head>
</html>
```

➔ Helps to target only a single webpage

->External: <link href="style.css"/>

```
<link rel="stylesheet" href="C:\Practice\vscode\CSS\styles.css">
```

➔ Helps to target website

Types of Selectors:

Element selector

```
body{
  background-color: #252525;
  margin:0;
}
hr{
  background-color: white;
  border-style:dotted none none;
  border-top-style:dotted;
  border-width:3px;
  height:0px;
  width:100%;
}
img:hover{
  background-color: white;
  height:50px;
}
h1{
  color:white;
  margin:0;
}
h3{
  color:white;
}
label{
  color:white;
}
a{
  color:white;
}
p{
  color:white;
  margin:10px 20px 20px 20px;
}
li{
  color:white;
}
```

```
td{
    color:white;
}
th{
    color:white;
}
div{
    background-color: black;
    margin-top: 0px;
}
```

Class Selector:

```
.circular{
    border-radius: 100%;
    height:50px;
}
```

ID selector:

```
#id{
    color:black;
}
*{
    Color:black; /*selects all
}
```

S6V39 - Font

Font Sizes

1px = 1/96th inch

1pt = 1/72nd inch

1em = 100% of parent

1rem = 100% of root

```
<!DOCTYPE html>

<head>
  <meta charset="UTF-8">
  <title>Font-Size</title>
  <style>
    #pixel {
      font-size: 20px;
    }

    #point {
      font-size: 20pt;
    }

    #em {
      font-size: 1em;
    }
  </style>
</head>
```

```

    #rem {
      font-size: 1rem;
    }

    footer {
      font-size: 12pt;
    }

    html {
      font-size: xx-large;
    }
  </style>
</head>
<html lang="en">

<body>
  <p id="pixel">1 Pixel is 1/96 of an Inch</p>
  <p id="point">1 Point is 1/72 of an Inch</p>
  <footer>
    <p id="em">1em is 100% the size of the parent element</p>
    <p id="rem">1rem is 100% the size of the root element</p>
  </footer>
</body>

</html>

```

Font Family:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Font Family</title>
  <style>
    h1 {
      font-family: 'Architects Daughter', cursive;
    }

    #helvetica {
      font-family: Helvetica, sans-serif;
    }

    #arial {
      font-family: Arial, sans-serif;
    }

    #serif {
      font-family: serif;
    }

    #sans-serif {
      font-family: sans-serif;
    }
  </style>

```

```

#cursive {
  font-family: cursive;
}

#monospace {
  font-family: monospace;
}

#fantasy {
  font-family: fantasy;
}
</style>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Architects+Daughter&display=swap"
rel="stylesheet">
</head>

<body>
  <h1>Font Family</h1>
  <p id="helvetica">Helvetica</p>
  <p id="arial">Arial</p>
  <p id="serif">Serif</p>
  <p id="sans-serif">Sans Serif</p>
  <p id="cursive">Cursive</p>
  <p id="monospace">Monospace</p>
  <p id="fantasy">Fantasy</p>
</body>
</html>

```

Font Weight:

Normal, Bold - keywords

Lighter, bolder - increase and decrease by value 100

Number - ranging 100-900

Text Align:

```

h1{
  text-align: start;
  text-align: end;
  text-align: center;
  text-align: justify;
}

```

S6V41 - Box Model

- Margin
- Border
- Padding
- Width & Height
- <div> Content Division </div>

S7V43 - Specificity and Inheritance

Position:

```
li{  
  color: red;  
  color: blue; //blue will be considered bcoz of latest position declaration  
}
```

Specificity:

```
li{color: blue;} //4 -element  
.class{color: red;} //3-class  
li[draggable]{color: green;} //2-attribute  
#id{color: black;} //1-id
```

Type:

```
<link rel="stylesheet" href="./solution.css"> //External  
<style> </style> //Internal  
<h1 style="">Hello</h1> //Inline
```

Importance:

```
#id{  
  color:black;  
  color:green !important;  
}
```

S7V45 - Positioning

Static: Default HTML Positioning

Relative: Position relative to default position

```
.blue-box {  
  background-color: blue;  
  width: 500px;  
  height: 300px;  
  position: relative;  
  top: 200px;  
  left: 200px;}
```

Absolute: Position relative to nearest positioned ancestor or top left corner of web page.

```
.red-circle {  
  width: 200px;  
  height: 200px;  
  background-color: red;  
  border-radius: 50%;  
  position: absolute;  
  top: 150px;  
  left: 250px;}
```

Fixed: Position related to top left corner of browser window & it won't change even if we scroll

S8V48 - CSS Display Properties

- **Block:** Height & width can change but only in same line
- **Inline:** Height and width can't change
- **Inline-Block:** Height and width can change & if width extends limit then needs a new line

S8V49 - CSS Float Properties

```
img{  
  float:left } // Newspaper column and inline pics adjustment
```

Incase if we need footer to set down and it is inline with image..
then to clear footer having float property

```
footer{  
  clear:left; }
```

S8V50 - Site Responsive - Adjust automatically with respect to size of screen

1-Media Queries:

```
@media screen {  
  //Inside CSS can be automatically adjusted by media screen query }
```

Min, max & combine

```
@media(min-width:600px) and (max-width:900px){  
  //styles for screens btwn 600px & 900px }
```

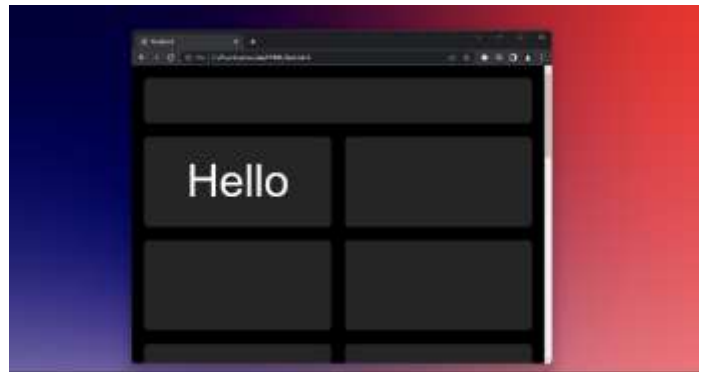
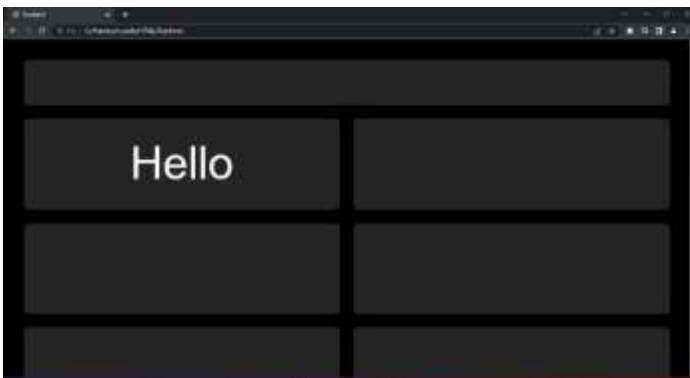
2-Grid:

```
<html>  
  <head>  
    <link rel="stylesheet" href="../CSS/test.css">  
  </head>  
  <body>  
    <div class="grid-container">  
      <div class="first card"></div>  
      <div class="card">  
        <p>Hello</p>  
      </div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
      <div class="card"></div>  
    </div>  
  </body>  
</html>
```

In CSS:

```
body{
  background-color: black;
  margin:3% 3% 3% 3%;
}
@media screen {
  p{
    color: white;
    font-family:sans-serif;
    font-size: 100px;
    text-align:center;
    margin: 0;
    margin-top: 40px;
  }
  .grid-container{
    display:grid;
    grid-template-columns: 1fr 1fr; //ratio of each column frame
    grid-template-rows: 100px 200px 200px 200px 200px 200px 200px 200px 200px 200px;
    gap: 30px; //gap btwn each row
  }
  .first{
    grid-column: span 2; //merging both columns in the first row
  }
  .card{
    background-color: #252525;
    border-radius: 10px;
  }
}
```

Automatic screen resolution was set by media screen query & CSS Grid implemented



S10V60: CSS Grid

Its mostly helpful for 2D alignment

```
display:grid;
grid-column: 400px 800px;
grid-row: 100px 200px;
//short hand notation
grid-template: 100px 200px / 400px 800px; // rows/columns
```

S10V61: Grid Sizing

```
.grid-container{
  display:grid;
  grid-template-rows: 100px auto; //automatically adjust to horizontal div
  grid-template-columns: 200px auto; //automatically adjust to horizontal div
}

.grid-container{
  display:grid;
  grid-template-rows: 1fr 2fr; // 1 ratio fraction and 2 ratio fraction
  grid-template-columns: 1fr 2fr;
}

.grid-container{
  display:grid;
  grid-template-rows: repeat(2, 200px); // will repeat 200px height for 2 times
  grid-template-columns: repeat(2, 100px); // will repeat 100px width for 2 times
  grid-auto-rows: 300px; // adjust the remaining undeclared divs to given size
}

.astronaut {
  background-color: #03989E;
  grid-area: 2 / 1 / 3 / 3; // row_start/column_start/row_end/column_end
}
```

3-CSS Flexbox:

Priority List: Content width < Width < flex-basis < min-width/max-width

Its mostly helpful for 1D alignment

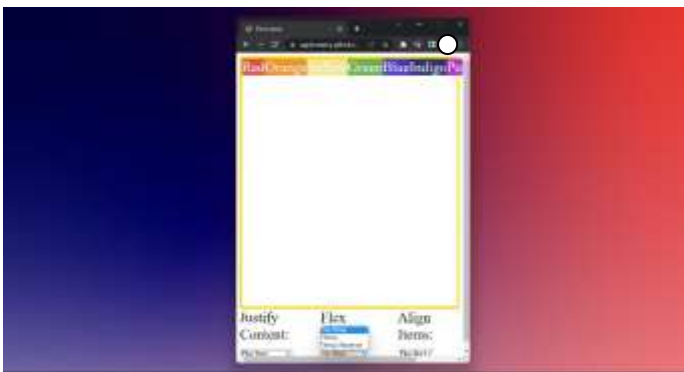
If we want to lay down all the divs in horizontal or in vertical then flex boxes are used.

- flex-direction: column; // vertical alignment
- flex-direction: row; //horizontal alignment
- flex-basis: 100px; // modifies the mail line size, which is perpendicular to cross line.
- order:1; //will changes the position of flex to right most if direction is in row mode.

S9V56- Flex Layout: <https://appbrewery.github.io/flex-layout/>

flex-wrap: nowrap;

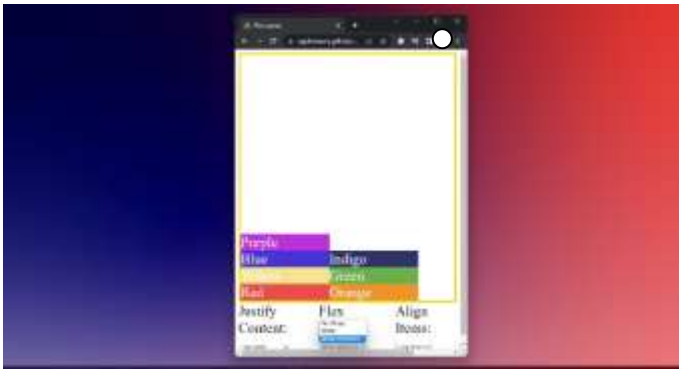
//default setting which doesn't adjust elements by screen size.



flex-wrap: wrap;

//respond to the screen adjustments and move elements. //parent property.

```
flex-wrap: wrap-reverse;  
//reverse the order of the flex alignment
```



```
justify-content: flex-start;  
// all the elements starts from left.
```

```
justify-content: flex-end;  
// all the elements starts from right.
```



```
justify-content: centre;  
// all the elements in centre justify mode.
```

```
justify-content: space-between;  
//Adjust uniform space between elements and corners stick to the end.
```



```
justify-content: space-around;  
//maintain spaces between elements and edges also.
```

```
justify-content: space-evenly;
```

```
// maintain space evenly to all elements including corner elements and edges.
```



```
align-items: flex-start;
```

```
//Align elements to the starting of the page
```

```
align-items: flex-end;
```



```
align-items: center;
```

```
//if we want to modify only a particular element from the items set.. then we can use
```

```
align-self: flex-start
```

```
// if we use this to particular violet.. then it will be sticked to the top and remaining  
will be set to the center. // to achieve particular height to be considered for centring  
the flex items then
```

```
height: 70vh; // here vh is viewpoint height
```

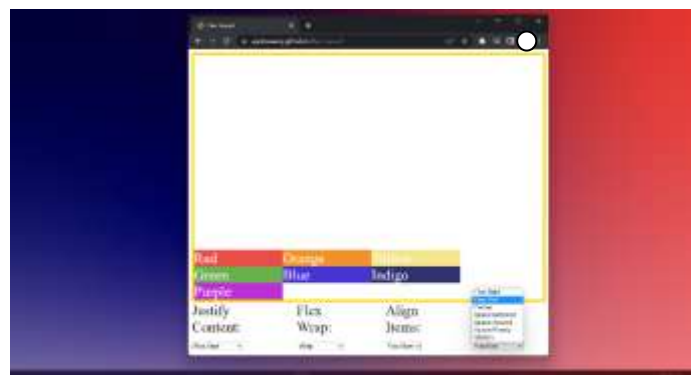
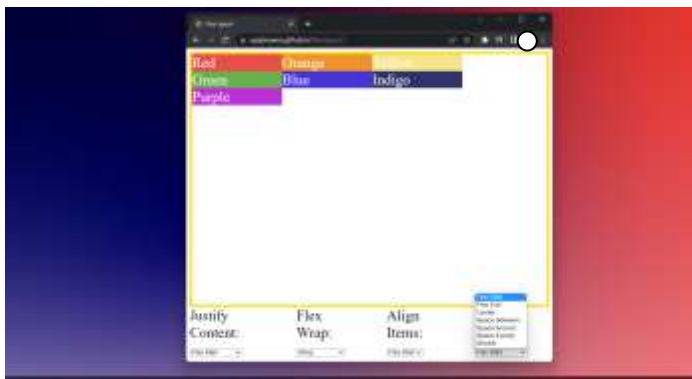
```
align-items: baseline;
```



```
align-items: stretch;
```

```
-> align-content is similar to align-items but only if the flex-wrap is set to wrap.
```

```
flex-wrap: wrap;  
align-content: flex-start;
```



```
flex-wrap: wrap;  
align-content: flex-end;
```

```
flex-wrap: wrap;  
align-content: center;
```



```
flex-wrap: wrap;  
align-content: space-between;
```

```
flex-wrap: wrap;  
align-content: space-around;
```



```
flex-wrap: wrap;  
align-content: space-evenly;
```

Exercise:

```
<html>
  <head>
    <link rel="stylesheet" href="../CSS/test.css">
  </head>
  <body>
    <div class="flex-container">
      <div class="first"></div>
      <div class="card"></div>
      <div class="card"></div>
      <div class="card"></div>
    </div>
  </body>
</html>
```

```
body{
  background-color: black;
  margin:3% 3% 3% 3%;
}
@media screen {
  .flex-container{
    display:flex;
  }
  .first{
    background-color: #252525;
    border: 5px solid white;
    height: 100px;
    flex: 2; //2 times the ratio of width
  }
  .card{
    background-color: #252525;
    border: 5px solid white;
    height: 100px;
    flex: 1; //ratio of width
  }
}
```



Priority List: Content width < Width < flex-basis < min-width/max-width

Flex Items :

```
flex-grow: 1;
flex-shrink: 1;
flex-basis: 100px;

//short form declaration
flex: 1 1 0; // 1st value grow, 2nd value shrink, 3rd value basis
flex: 1; //grow=1, shrink=1, basis=1;
```

4-Bootstrap: [External Framework] 5D Type of CSS framework

//width is divided into 12 columns which is used as the default column navigating style

- Most popular widely used CSS type in world
- 12 column layout system will help more to make
- site responsive
- Include via CDN (Content Delivery Network)
- Custom style should be placed below the CDN link to get its priority

Breakpoint	Class infix	Dimensions	Devices
Extra small	None	<576px	Thin screens
Small	sm	>=576px	Mobile
Medium	md	>=768px	IPads
Large	lg	>=992px	Laptops
Extra Large	xl	>=1200px	Desktops
Extra extra Large	xxl	>=1400px	TV

Multiple break points declaration

```
<div class="col-sm-12 col-md-8 col-lg-4">CTS</div>
```

- On small devices take 12/12
- On medium devices take 8/12
- On large devices take 4/12

S11V66: DarkMode 27:16

```
<html lang="en" data-bs-theme="dark">
```

Dark mode applicable depends on the system default theme.

```
@media (prefers-color-scheme:light){
  body{
    background-color: white;
  }
  h1{
    color:black;
  }
}
@media (prefers-color-scheme:dark){
  body{
    background-color: black;
  }
  h1{
    color:white;
  }
}
```

S12V69: Understand color theory

S12V70: Typography

S12V71: UserInterface(UI)

S12V72: User Experience(UX)

- Simplicity
- Consistency

S14V76: Intro to Java script

- > The language that powers the web
- > JavaScript is an interpreted programming language
- > Java is a compiled programming language

Interpreted	Compiled
• JavaScript • Python • Ruby	• Java • C/C++ • Swift



S14V77:

In chrome → more tools → developer tools → source tab → index.js →

```
alert("Hello");
```

press ctrl+Enter will create popup alert message in browser

- alert- function
- Hello- Message
- ;- End

S14V77: MDN Website about JS web docs resources

<https://developer.mozilla.org/en-US/docs/Web/JavaScript>

Idiomatic Javascript : <https://github.com/rwaldron/idiomatic.js>

S14V78: Data types

- `typeof("Hello")` → String
- `typeof(23)` → Number
- `typeof(true)` → boolean

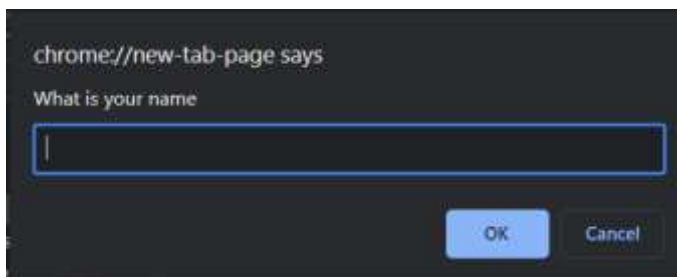
S14V79: Variables

```
var myName = "VR"
```

In console type myName... It will give stored VR name from variable (or) `alert(myName)`

- Var - keyword
- myName - Name
- VR - Value

```
var yourName = prompt("What is your Name : ?")
```



In console type yourName → it gives the name what we entered

```
alert("My Name is " + myname + " Welcome " + yourName)
```

To clear console press Ctrl+L

S14V80: JavaScript variables exercise start

Instructions

Javascript Variables Exercise

Given the existing code below, can you write some code so that their values are switched around?

```
var a = "3";  
var b = "8";
```

So that the variable `a` holds the value "8".
And the variable `b` holds the value "3".
When the code is run, it should output:

```
a is 8  
b is 3
```

Do **NOT** change any of the existing code.
You are **NOT** allowed to type any numbers.
You should **NOT** redeclare the variables `a` and `b`.
Hint: Use [this code playground](#) to run your code and see if it matches your expectations.
Hint: The solution is just 3 lines of code.

index.js

```
1 function test() {  
2   var a = "3";  
3   var b = "8";  
4  
5   /*****Do not change the code above *****/  
6   //Write your code on lines 7 - 9:  
7   var temp;  
8   temp=a;  
9   a=b;  
10  b=temp;  
11  
12  
13  /*****Do not change the code below *****/  
14  
15  console.log("a is " + a);  
16  console.log("b is " + b);  
17 }  
18
```

Run tests Run code

All changes saved | Line 18, Column 1

Result Success

Test Cases Failed: 0, Passed: 1 of 1 tests

Test result User logs

Test a should be 8 and b shc

Your code passed this test

Coding Exercise

Next

Naming of variables: CamelCasing

```
1 var myName = "Angela";  
2 var yourName = "Jack Bauer";  
3 var myvar = "abc";  
4 var my123 = 123;  
5 var myname = "Angela";  
6 var my_name = "Angela";  
7 var userScoreFinal = 12;  
8
```

S14V83: String Concatenation

```
1 var message = "Hello";  
2 var name = "Angela";  
3  
4 //Write your code below this line:  
5  
6 alert(message + " there," + " " + name);
```

S14V84 : String Lengths and Retrieving the Number of Characters.

chrome://new-tab-page says

You have entered Hello with 5 characters and you left 175 remaining!

OK

index.js

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10 var tweet = prompt("Enter Text");  
11 alert("You have entered " + tweet + " with " + tweet.length + " characters and you left " + (180-tweet.length) + " remaining!");  
12
```

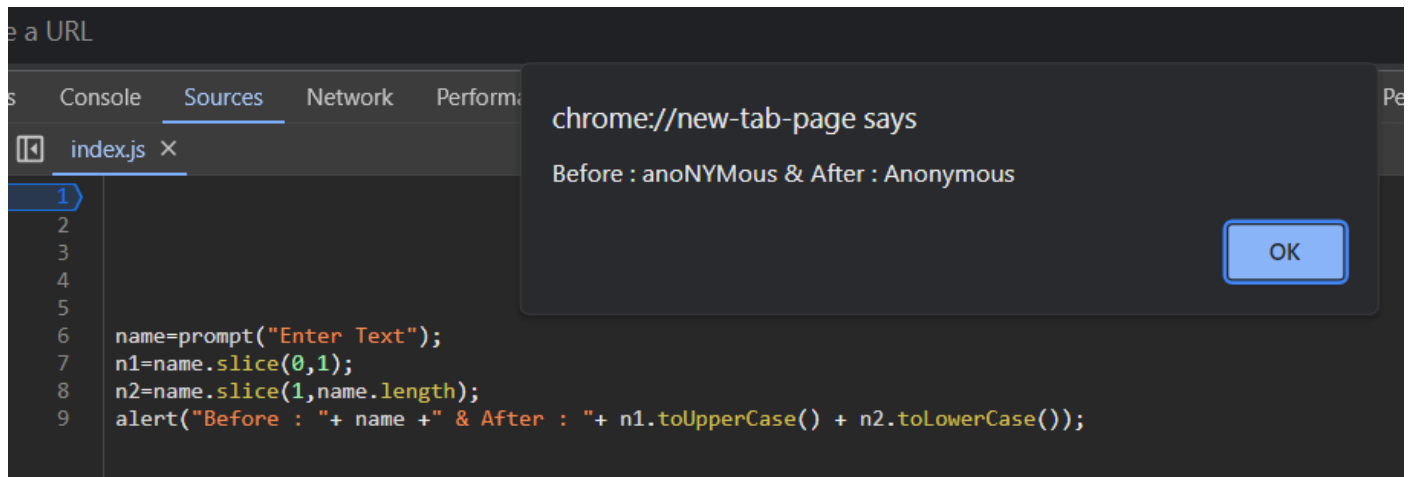
S14V85: Slicing Characters.

```
var name = "Angela";  
name.slice(1,5);
```

Answer will be nge
starts from index 1-n and ends before index 5-a

S14V86: Uppercase and lowercase... Changing Case of the string

```
name = "Anonymous";  
name1 = name.toUpperCase();  
name2 = name.toLowerCase();
```



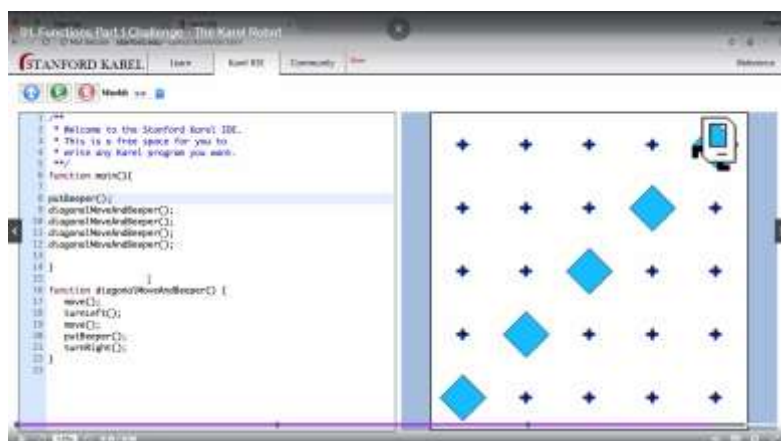
S14V88: Basic Arithmetic and Modulo Operator in JavaScript

> 2+3	Addition
< 3	
> 5-2	Subtraction
< 3	
> 2*3	Multiplication
< 6	
> 7/4	Division
< 1.75	
> 6%4	Modular : Gives Remainder
< 2	

S14V89: Incremental and Decremental Expressions

```
1 var n=7;  
2 n++; //7+1=8  
3 n-=5; //8-5=3
```

S14V90: Functions Part1



S14V94: Functions Part2

```
index.js x
1 function arithmetic(x,y){
2   //var x=prompt("Enter x: ");
3   //var y=prompt("Enter y: ");
4   var add=x+y;
5   var sub=x-y;
6   var mul=x*y;
7   var div=x/y;
8   var mod=x%y;
9   console.log(add+" "+sub+" "+mul+" "+div+" "+mod);
10 }
11
12 arithmetic(4,2); //6 2 8 2 0
```

Life in Weeks Quiz:

Instructions

LIFE IN WEEKS Coding Exercise

I was reading this article by [Tim Urban - Your Life in Weeks](#) and realised just how little time we actually have.

In this challenge, you are going to create a function that tells us how many days, weeks and months we have left if we live until 90 years old.

It will take your current age as the input and console.logs a message with our time left in this format:

You have x days, y weeks, and z months left.

Where x, y and z are replaced with the actual calculated numbers.

For this challenge, assume there are **365** days in a year, **52** weeks in a year and **12** months in a year.

IMPORTANT your `console.log` output should match the Example Output format exactly, even the positions of the commas and full stops.

Example Input

e.g. If you are 55 years old:

```
lifeInWeeks(55)
```

Example Output

```
You have 12435 days, 2755 weeks, and 488 months left.
```

Hint

1. Try using [this Repl.it playground](#) to test your code and see if the output is what you expect it to be.
2. Make sure your console log output matches the example output precisely. The same capitalisation, the same spaces, commas and full stops.

index.js

```
1 function lifeInWeeks(age) {
2
3   /*****Don't change the code above*****/
4
5   var remAge = (90-age);
6   console.log("You have "+(remAge*365)+" days, "+(remAge*52)+" weeks, and "+(remAge*12)+" months left.");
7
8   /*****Don't change the code below*****/
9 }
10
11
```

Run tests Run code

All changes saved | Line 11, Column 1

Result **Success**

Test Cases

Failed: 0, Passed: 3 of 3 tests

- ✓ lifeInWeeks() Problem! The correct solution should be "You have 14235 days, 2025 weeks, and 468 months left." When age is 51.
- ✓ lifeInWeeks() Problem! The correct solution should be "You have 22630 days, 3234 weeks, and 744 months left." When age is 28.
- ✓ lifeInWeeks() Problem! The correct solution should be "You have 18425 days, 2340 weeks, and 540 months left." When age is 40.

Next

S14V96: Functions Part 3 - Outputs and Return Values

Creating the function

```
function getMilk (money) {
  return money % 1.5;
}
```

Calling the function

```
var change = getMilk(4);
```

The App Brewery

A return statement ends the execution of a function, and returns control to the calling function.

Execution resumes in the calling function at the point immediately following the call.

A return statement can return a value to the calling function

```
index.js x
1
2
3
4
5 function getMilk(money){
6   alert("Buy "+calcBottles(money,1.5)+" bottles and get Change $"+(money*1.5));
7   return money*1.5; //Remainder of div
8 }
9
10 function calcBottles(startMoney, costPerBottle){
11   var noOfBottles=Math.floor(startMoney/costPerBottle);
12   return noOfBottles;
13 }
14
15 getMilk(5);
```

-> calcBottles function returns noOfBottles to getMilk function

-> getMilk function returns change money in console log

Coding Exercise:

BMI Calculator Challenge

Create a BMI calculator using JavaScript functions.

The Body Mass Index (BMI) is a way of estimating the amount of body fat. It's used in medicine to calculate risk of heart disease.

You can calculate it using the formula below, where weight divided by height squared.

$$BMI = \frac{weight(kg)}{height^2(m^2)}$$

Your challenge is to create a function that takes weight and height as inputs and gives the calculated BMI value as an **output**. The output should be rounded to the nearest whole number.

The first parameter should be the weight and the second should be the height.

NOTE: You do not need to write any **alerts** or **prompts** or **console logs**. Your code should only contain the function, the result has to be **returned** by the function. You **do not** need to call the function.

[Hint](#)

```

1  /* If my weight is 60kg and my height is 1.8m, I should be able to call your function like this!
2  var bmi = bmiCalculator(60, 1.8);
3
4  bmi should equal 20 when it's rounded to the nearest whole number.
5
6  */
7
8  function bmiCalculator(weight,height){
9      var BMI = Math.round(weight/(height*height));
10     return bmi;
11 }
12 var bmi = bmiCalculator(60, 1.8);

```

Run tests Run code

All changes saved | Line 9, Column 22

Result Success

Test Cases
Failed: 0, Passed: 2 of 2 tests

Test result User logs

✓ Your code passed this test.

✓ bmiCalculator() ERROR: Your solution code should return the bmi as an output.

✓ bmiCalculator() ERROR: Your solution should round the number to the nearest whole number.

Next

S15V100: Random Number Generator using JavaScript

`Math.random();` //this will generate random numbers from 0 to 0.999(upto 16 decimal places)

index.js*

×

```

1  var n = Math.random();
2  n = n * 6;
3  n = Math.floor(n) + 1;
4  console.log(n);

```

//Dice random number generator code from 1 to 6

S15V101: Control statements - If Else

S15V102: Comparators and Equality

`===` Three equal comparator also checks the Datatype compared to `==` Two equal comparator

- `&&` - AND
- `||` - OR
- `!` - NOT

Coding Exercise:

BMI Calculator Advanced (IF/ELSE)

Previously, we've created a function that is able to calculate the BMI. But once we get a result, we will want to tell the user what the number means.

Write a function that outputs (returns) a different message depending on the BMI.

BMI <18.5, the output should be: "Your BMI is <bmi>, so you are underweight."

BMI 18.5-24.9, the output should be: "Your BMI is <bmi>, so you have a normal weight."

BMI >24.9, the output should be: "Your BMI is <bmi>, so you are overweight."

The message **MUST** be returned as an output from your function. You should **NOT** NEED to use **alert**, **prompt** or **console.log** in this challenge.

IMPORTANT the message wording has to match precisely for the code to pass the validation. Including punctuation and capitalisation.

BODY MASS INDEX



```

1  function bmiCalculator (weight, height) {
2      var BMI=weight/(height*height);
3      var interpretation;
4      if(BMI<18.5){
5          interpretation="Your BMI is "+BMI+" , so you are underweight.";
6      }
7      else if(BMI>=18.5 && BMI<=24.9){
8          interpretation="Your BMI is "+BMI+" , so you have a normal weight.";
9      }
10     else{
11         interpretation="Your BMI is "+BMI+" , so you are overweight.";
12     }
13     return interpretation;
14 }
15

```

Run tests Run code

All changes saved | Line 3, Column 24

Result Success

Test Cases
Failed: 0, Passed: 3 of 3 tests

Test result User logs

✓ Your code passed this test.

✓ bmiCalculator() should return "Your BMI is 16, so you are underweight." for 60, 2

✓ bmiCalculator() should return "Your BMI is 20, so you have a normal weight." for 80, 2

✓ bmiCalculator() should return underweight for 100, 2

Next

Code Challenge: Leap year

Instructions

Leap Year Challenge Exercise

👉 This is a Difficult Challenge 👈

Write a program that works out whether if a given year is a leap year. A normal year has 365 days, leap years have 366, with an extra day in February. The reason why we have leap years is really fascinating. [this video](#) goes into more detail.

This is how to work out whether if a particular year is a leap year:

A year is a leap year if it is evenly divisible by 4, **except** if that year is also evenly divisible by 100, **unless** that year is also evenly divisible by 400.

e.g. Is the year 2000 a leap year?

$2000 \div 4 = 500$ (Leap)

$2000 \div 100 = 20$ (Not Leap)

$2000 \div 400 = 5$ (Leap)

So the year 2000 is a leap year.

But the year 2100 is not a leap year because:

$2100 \div 4 = 525$ (Leap)

$2100 \div 100 = 21$ (Not Leap)

$2100 \div 400 = 5.25$ (Not Leap)

Warning your output should match the Example Output format exactly, even the positions of the

Coding Exercise

index.js

```
1 function isLeap(year) {
2   if(year%4===0){
3     if(year%100===0){
4       if(year%400===0){
5         return("Leap year.");
6       }
7     }
8     else{
9       return("Not leap year.");
10    }
11  }
12  else{
13    return("Leap year.");
14  }
15 }
16 else{
17   return("Not leap year.");
18 }
```

Run tests Run code All changes saved | Line 18, Column 1

Result

Success at

Test Cases

Failed: 0, Passed: 3 of 3 tests

- ✓ isLeap() Problem! isLeap(1948) should return "Leap year."
- ✓ isLeap() Problem! isLeap(1968) should return "Not leap year."
- ✓ isLeap() Problem! isLeap(2020) should return "Leap year."

Test result

✓ Your code passed this test

Next

S15V106: Collections - Arrays

Console Sources Network Performance

index.js

```
1
2
3
4
5
6 var guestList = ["Anonymous","VR","John"];
7 guest-prompt("Enter Guest Name: ");
8 if(guestList.includes(guest)){
9   alert("Welcome "+guest);
10 }else{
11   alert("Better Luck Next Time");
12 }
```

chrome://new-tab-page says

Welcome VR

OK

Code Challenge: FizzBuzz

Elements Console Sources Network Performance Memory Application Security Lighthouse Recorder Performance insights

index.js

```
1 var output=[];
2 var n=1;
3 function fizzbuzz(){
4   if(n%3===0 && n%5===0){
5     output.push("FizzBuzz");
6   }else if(n%3===0){
7     output.push("Fizz");
8   }else if(n%5===0){
9     output.push("Buzz");
10  }else{
11    output.push(n);
12  }
13  n++;
14  console.log(output);
15 }
16 fizzbuzz();
17
18 // var n=1;
19 // for(n=1;n<=100;n++){
20 //   if(n%3===0 && n%5===0){
21 //     console.log("FizzBuzz");
22 //   }else if(n%3===0){
23 //     console.log("Fizz");
24 //   }else if(n%5===0){
25 //     console.log("Buzz");
26 //   }else{
27 //     console.log(n);
28 //   }
29 // }
```

Line 11, Column 23

Console

What's New

top | Filter

▶ (29) 1, 2, "Fizz", 4, "Buzz", "Fizz", 7, 8, "Fizz", "Buzz", 11, "Fizz", 13, 14, "FizzBuzz", 16, 17, "Fizz", 19, "Buzz", "Fizz", 22, 23, "Fizz", "Buzz", 26, "Fizz", 28, 29

▶ FizzBuzz()

▶ (30) 1, 2, "Fizz", 4, "Buzz", "Fizz", 7, 8, "Fizz", "Buzz", 11, "Fizz", 13, 14, "FizzBuzz", 16, 17, "Fizz", 19, "Buzz", "Fizz", 22, 23, "Fizz", "Buzz", 26, "Fizz", 28, 29, "FizzBuzz"

Code Challenge: Who's Buying Lunch

Who's Buying Lunch? Code Challenge

You are going to write a function which will select a random name from a list of names. The person selected will have to pay for everybody's food bill.

Important: The output should be returned from the function and you do not need `alert`, `prompt` or `console.log`. The output should match the example output exactly, including capitalisation and punctuation.

Example Input

```
["Angela", "Ben", "Jenny", "Michael", "Chloe"]
```

Example Output

```
Michael is going to buy lunch today!
```

Hint

1. You might need to think about [Array.length](#)
2. Remember that Arrays start at position 0!

```
1 function who'sPaying(names) {
2
3 //*****Don't change the code above*****/
4
5 var myList=["Angela", "Ben", "Jenny", "Michael", "Chloe"];
6 var index=Math.round(Math.random()*myList.length);
7 return(myList[index]+" is going to buy lunch today!");
8
9 //*****Don't change the code below*****/
10 }
```

Run tests Run code All changes saved | Line 8, Column 1

Result Success

Test Cases
Failed: 0, Passed: 2 of 2 tests

- ✓ who'sPaying() The last person in the names Array is never picked.
- ✓ who'sPaying() The first person in the names Array is never picked.

Test result Your code passed this test

Coding Exercise

Next

S15V109: Control Statements - While loop

Console Sources Network Perform

index.js X

```
1 var output=[];
2 var n=1;
3 function fizzbuzz(){
4   while(n<=15){
5     if(n%3===0 && n%5===0){
6       output.push("FizzBuzz");
7     }else if(n%3===0){
8       output.push("Fizz");
9     }else if(n%5===0){
10      output.push("Buzz");
11    }else{
12      output.push(n);
13    }
14    n++;
15    console.log(output);
16    alert(output);
17  }
18  fizzbuzz();
19 }
```

chrome://new-tab-page says
1,2,Fizz,4,Buzz,Fizz,7,Fizz,Buzz,11,Fizz,13,14,FizzBuzz

OK

Code Challenge: Fibonacci Generator

The Fibonacci Exercise

Fibonacci was an Italian mathematician who came up with the [Fibonacci sequence](#):
0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 ...

Where every number is the sum of the two previous ones.
e.g. 0, 1, 1, 2, 3, 5 comes from

```
0 + 1 = 1
1 + 1 = 2
1 + 2 = 3
2 + 3 = 5
etc.
```

Create a function where you can call it by writing the code:
`fibonacciGenerator(n)`

Where `n` is the **number of items** in the sequence.
So I should be able to call:
`fibonacciGenerator(1)` and get
`[0,1]`
as the output.

IMPORTANT: The solution checker is expecting an **array** as the correct output.
Do **NOT** change any of the existing code.

You do **NOT** need any **alerts** or **prompts**, the result should be **returned** from the function as an **output**.

The **first two numbers** in the sequence must be 0 and 1.

Also, if you decide to create a **for** loop, make sure you explicitly specify `var i = 0` rather than simply writing `i = 0`. This is a quirk of the testing suite.
e.g. `for (var i = 0; i < 10; i++)`.

```
1 function fibonacciGenerator (n) {
2 //Do NOT change any of the code above
3   var output=[];
4   if(n===1){
5     output=[0];
6   }else if(n===2){
7     output=[0,1];
8   }else{
9     output=[0,1];
10    for(var i=2;i<=n;i++){
11      output.push(output[output.length-2]+output[output.length-1]);
12    }
13  }
14  //Return an array of fibonacci numbers starting from 0.
15  return(output);
16  //Do NOT change any of the code below
17 }
18
19 }
```

Run tests Run code All changes saved | Line 7, Column 5

Result Success

Test Cases
Failed: 0, Passed: 3 of 3 tests

- ✓ fibonacciGenerator() Problem! The correct solution should return [0] as the output when fibonacciGenerator(1) is called
- ✓ fibonacciGenerator() Problem! The correct solution should return [0, 1] as the output when fibonacciGenerator(2) is called
- ✓ fibonacciGenerator() Problem! The correct solution should return [0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377] as the output when fibonacciGenerator(15) is called

Coding Exercise

Next

S16V115: Adding JavaScript to websites

- Inline JS
- Internal JS
- External JS

Inline JS

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
  <meta charset="UTF-8">
  <title>DOM Website</title>
  <link rel="stylesheet" href="../DOM/style.css">
</head>
<body onload="alert('Hello');">
  <h1>Hello</h1>
</body>
</html>
```

Internal Js

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
  <meta charset="UTF-8">
  <title>DOM Website</title>
  <link rel="stylesheet" href="../DOM/style.css">
</head>
<body>
  <h1>Hello</h1>
  <script type="text/javascript">
    alert("Hello");
  </script>
</body>
</html>
```

External JS:

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
  <meta charset="UTF-8">
  <title>DOM Website</title>
  <link rel="stylesheet" href="../DOM/style.css">
</head>
<body>
  <h1>Hello</h1>
  <script src="../DOM/index.js" charset="UTF-8"></script>
</body>
</html>
```

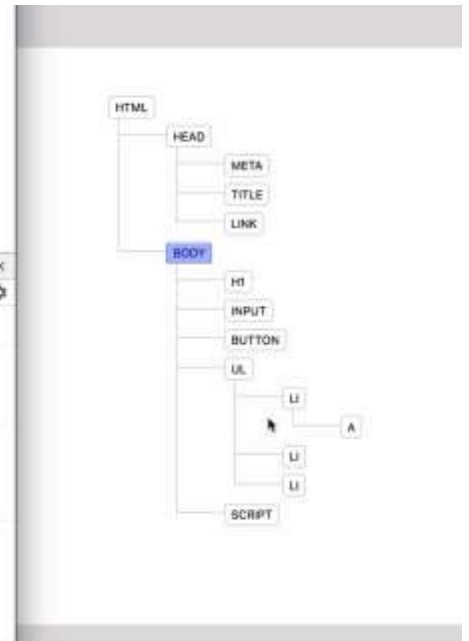
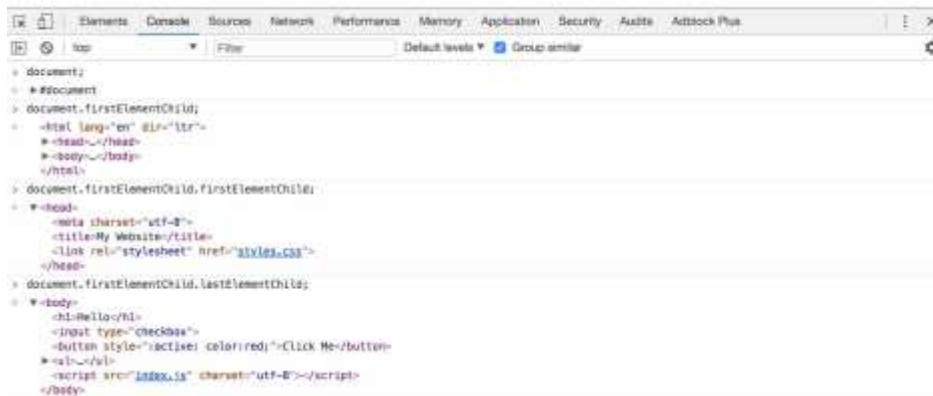
//always place javascript at the bottom near to </body> tag bcoz java script is to change the existing code.. so need to take some time to load the upper part of the code.

S16V116: Document Object Model (DOM)

The Document Object Model is a cross-platform and language-independent interface that treats an HTML or XML document as a tree structure wherein each node is an object representing a part of the document. The DOM represents a document with a logical tree.

Hello

- ☐ Click Me
- Google
- Second
- Third



```
document.getElementsByTagName("btn")[0]; //As mentioned getElements, it will gather all
elements with tagname btn..need to mention index to get specific element
document.getElementsByClassName("li")[2]; //index 2 for 3rd element in the list
document.getElementById("title"); // title ID
document.querySelector("h1").innerHTML="Hello World" // Changing H1's innerHTML text to
hellow world using querySelector
document.querySelectorAll("h1").innerHTML="Hello World" // Changes all H1 tags innerHTML
text to hellow world using querySelector. For specific element we need to add [2] index
```

CSS

- visibility: hidden; // separated with “-“
- background-color : red;

JS

- visibility = “hidden”; // CamelCasing
- backgroundColor = “red”;

S16V120: Separation of Concerns

```
> document.querySelector("button").classList.add("invisible");
undefined
> document.querySelector("button").classList.remove("invisible");
undefined
> document.querySelector("button").classList.toggle("invisible");
true
> document.querySelector("button").classList.toggle("invisible");
false
```

If a particular element is not looking in the way it want to be, then we can simply dig into the stylesheet

If a particular element is not behaving in the way it want it to, then we need to do changes in javascript.

S16V121: Text Manipulation and Text Content Property

```
document.querySelector("h1").innerHTML("<em>Hello</em>");// need to place tags also in  
between "", so that JS will consider <em> also
```

S16V122: Manipulating HTML Element Attributes

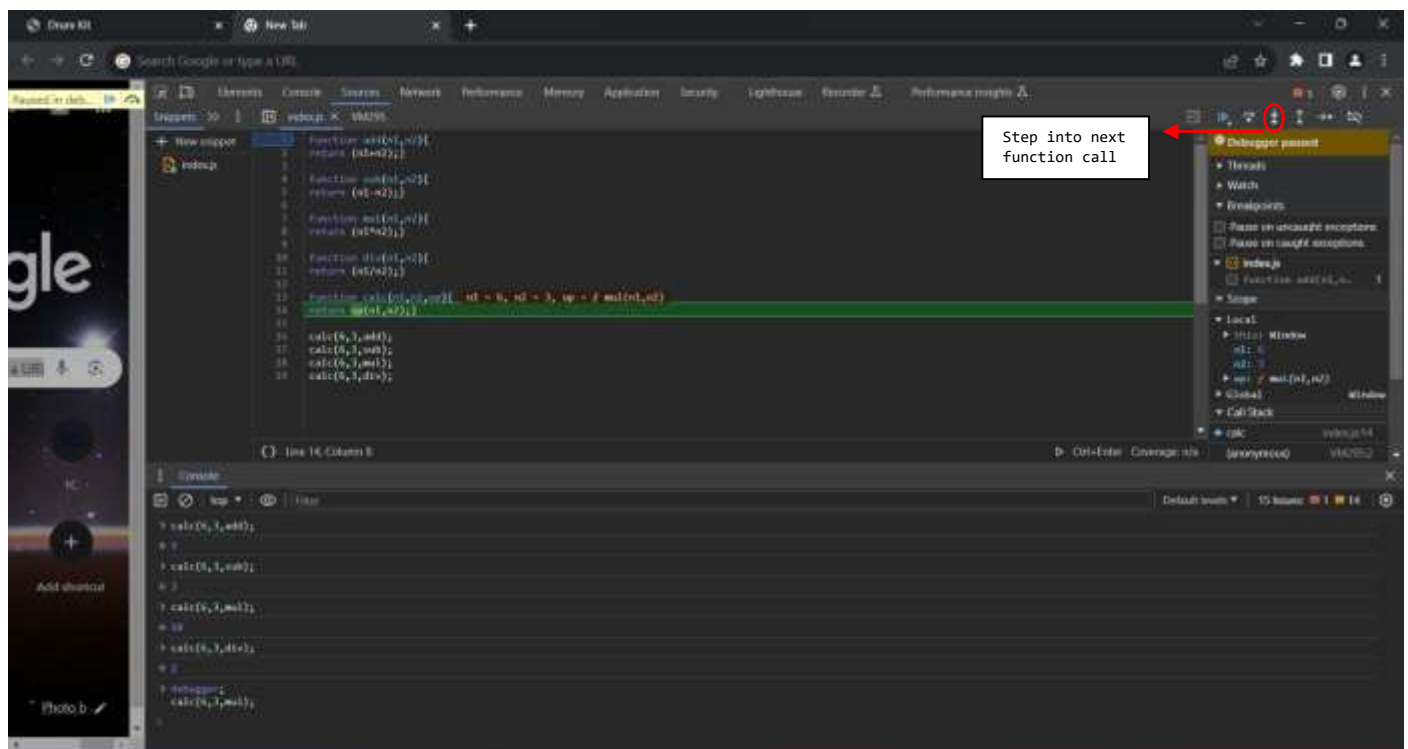
```
document.querySelector("a").attributes; //will give all the list of attributes attached to  
HTML element  
document.querySelector("a").getAttribute("href"); //will give the value given in  
particular attribute  
document.querySelector("a").setAttribute("href","https://bing.com"); //will change old  
link to new given link ("attribute","new value")
```

In JS:

```
document.querySelectorAll("button").addEventListener("click",handleClick);  
//After click function the function should be called  
//If we add handleClick() along with parenthesis.. then the method will call straight up  
instead of waiting for click method.  
//Instead of calling the function by handleClick().. we are passing the method by  
handleClick only after the action performed.
```

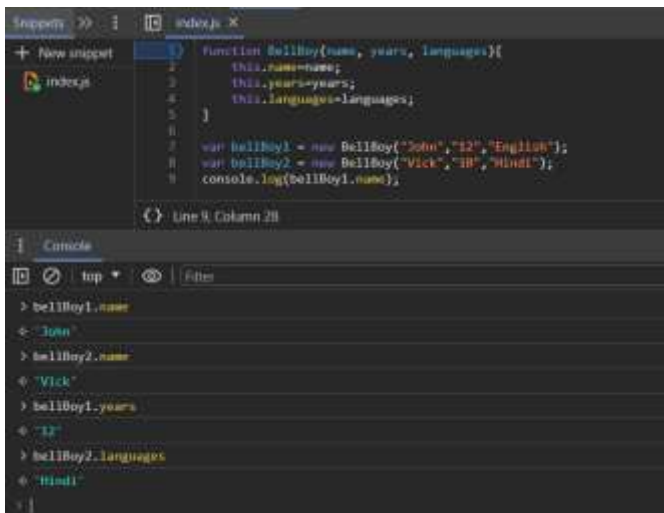
```
function handleClick(){  
    alert("Clicked");  
}
```

Calling function in another function : Calculator



Debugging helps to navigate step by step flow of the execution.

S18V141: Constructor Function



The screenshot shows a VS Code editor with a snippet named 'index.js'. The code defines a constructor function 'BellBoy' with parameters 'name', 'years', and 'languages'. It initializes 'this.name', 'this.years', and 'this.languages'. Then, it creates two instances: 'bellBoy1' with 'John', '12', and 'English', and 'bellBoy2' with 'Vick', '10', and 'Hindi'. The console output shows the names and languages of both instances.

```
function BellBoy(name, years, languages){
  this.name=name;
  this.years=years;
  this.languages=languages;
}

var bellBoy1 = new BellBoy("John","12","English");
var bellBoy2 = new BellBoy("Vick","10","Hindi");
console.log(bellBoy1.name);
```

Console output:

```
> bellBoy1.name
"John"
> bellBoy2.name
"Vick"
> bellBoy1.years
"12"
> bellBoy2.languages
"Hindi"
```

S18V142: Switch statements in JS

```
switch(buttonClick){
  case("w"):
    var audioW = new Audio("sounds/tom-1.mp3");
    audioW.play();
    break;
  case("a"):
    var audioA = new Audio("sounds/tom-2.mp3");
    audioA.play();
    break;
  default:console.log();
}
```

S18V143: Objects their Methods and Dot notation - Constructor Function

- A constructor is a special function that creates and initializes an object instance of a class.
- In JavaScript, a constructor gets called when an object is created using the new keyword.
- The purpose of a constructor is to create a new object and set values for any existing object properties

```
function BellBoy(name, years, languages){
  this.name=name;
  this.years=years;
  this.languages=languages;
  this.cleaning = function(){ //creating a new variable and constructing an anonymous function.
    alert("Cleaning in Progress");
  }
}

var bellBoy1 = new BellBoy("John","12","English");
var bellBoy2 = new BellBoy("Vick","10","Hindi");
bellBoy1.cleaning();
```

S18V145: Keyboard keypress events

Keyboard Events

Event Name	Fired When
keydown	ANY key is pressed
keypress	ANY key except Shift, Fn, CapsLock is in pressed position. (Fired continuously.)
keyup	ANY key is released

S18V146: Higher Order Function and Callback Function

```
document.addEventListener("keydown",respondToKey(event){
    alert("Key Pressed");
})
```

- addEventListener – Higher Order Function
- respondToKey – Callback Function

S18V147: Adding Animations to Website

S19V150: JQuery

- jQuery is a JavaScript library designed to simplify HTML DOM tree traversal and manipulation, as well as event handling, CSS animations, and Ajax.
- It is free, open-source software using the permissive MIT License.
- CDN (Content Delivery Network) for JQuery:
https://www.w3schools.com/jquery/jquery_get_started.asp

```
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<script src="index.js" charset="utf-8"></script> // Always add JQuery CDN above the JS
```

```
$(document).ready(function(){ // To check whether JQuery is loaded completely and ready
    $("h1").css("color","red");
})
```

S119V152: Minification work to reduce size <https://www.minifier.org/>

S119V154-156: Manipulating with JQuery

```
$("h1").hasClass("margin-50"); //manipulates styles
true

$("h1").text("Good BYE"); //Manipulates Text
$("button").html("<em>Hello</em>");//Manipulates html also

$("a").attr("href","www.google.com");//Attribute Manipulation
```

S19V157: Adding EventListener

```
for(var i=0;i<5;i++){
    document.querySelectorAll("button")[i].addEventListener("click",function(){
        document.querySelector("h1").style.color = "purple";
    })//using for loop we will access all the buttons
}

$("button").click(function(){
    $("h1").css("color","purple");// no for loop needed and syntax is diff
```

```
})
```

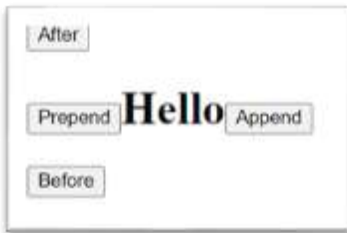
```
$("#input").keypress(function(evt){  
    console.log(evt.key); //eventlistener for keypress  
})
```

```
$(document).keydown(function(event){  
    $("#h1").text(event.key); //changing h1 text from keyboard input  
    console.log(event.key);  
})
```

```
$("#h1").on("mouseover",function(){ //”on” eventlistener  
    $("#h1").css("color","red");  
})
```

S19V157: Adding and Removing Elements with JQuery

```
$("#h1").after("<button>Before</button>");//after the h1 tag  
$("#h1").before("<button>After</button>");//before the h1 tag  
$("#h1").prepend("<button>Prepend</button>");//add in h1 tag just before original data  
$("#h1").append("<button>Append</button>");// add in h1 tag just after original data
```



S19V159: Add animations with JQuery

```
$("#button").on("click",function(){  
    $("#h1").hide();  
    $("#h1").show();  
    $("#h1").slideUp();  
    $("#h1").slideDown();  
    $("#h1").fadeIn();  
    $("#h1").fadeOut();//& numeric animations are not allowed in below line  
    $("#h1").slideUp().slideDown().animate({opacity:0.5}); //chaining animations 1 by 1  
})
```

S21V185: Downloading Git and setup

Download Git <https://gitforwindows.org/>

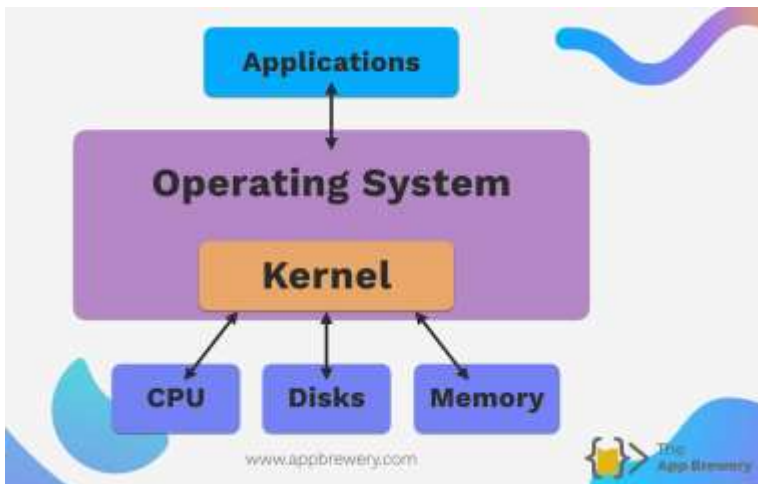
After downloading check in VSCode by

Terminal-> new terminal-> select + dropdown and check GitBash

To set GitBash as default

View-> CommandPallet-> Select default profile-> GitBash.

S21V186: Command Line



With the help of kernel we can able to access Hardware and software interface.

Graphic User Interface Shell (GUI)

Command line Interface Shell (CLI)

BASH (Bourne Again Shell) is a CLI for UNIX base Systems

S21V187: Navigation Directory

cd - Change Directory

cd .. - navigates 1 directory backwards

Press Ctrl+a to move cursor to starting of line in terminal

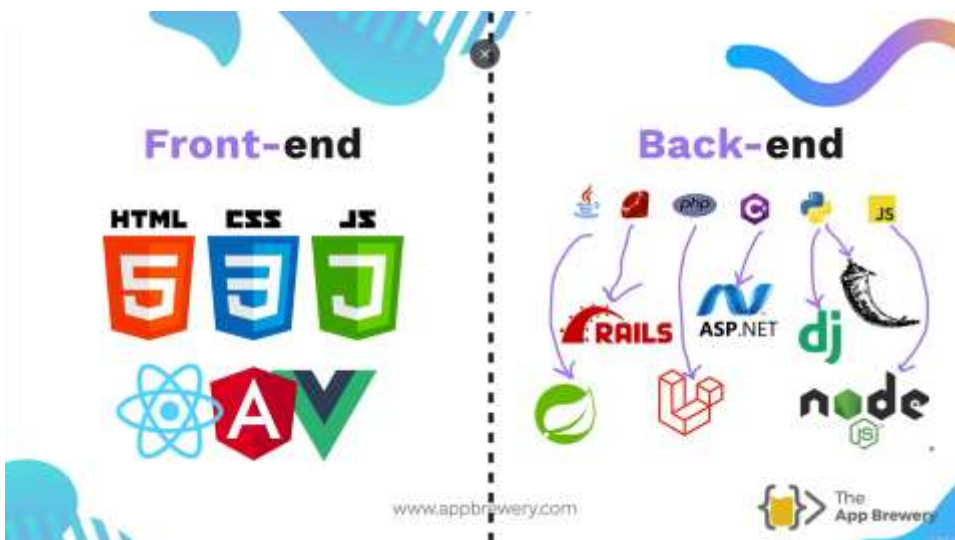
Press Ctrl+e to move cursor to end of line in Terminal

Press Ctrl+u to clear existing line in terminal.

S22V190: Backend

Consists

- Server - nothing but computer
- Application - code or script
- DataBase - Users Data



FrontEnd:

For Skeleton - HTML

For Styling - CSS

For Functionality - JavaScript

Frameworks - React, Angular, Vue

Backend:

Java - Spring

Ruby - Rails

Csharp - ASP.Net from Microsoft

php - Laravel

python - Django, flask

javascript - node.js

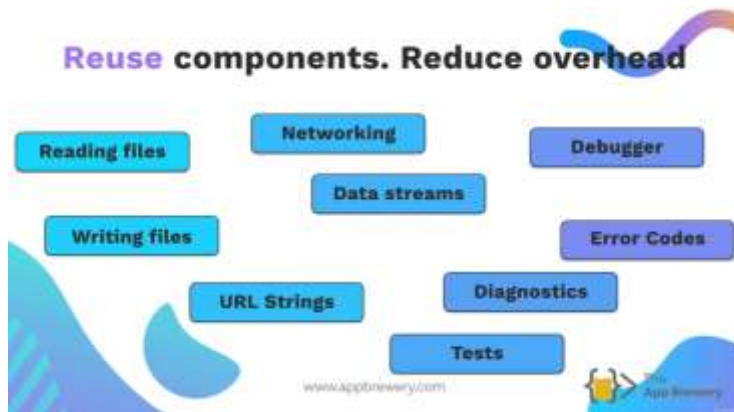
S23:Node.js

Node.js is an asynchronous event-driven JavaScript runtime and is designed to build scalable network applications -Node.Js Team

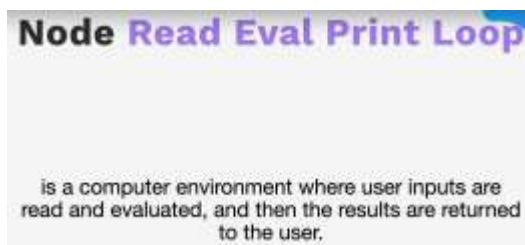
JavaScript runtime: In easy words, Node.Js helps to use Js outside the browser in local.

Asynchronous event-driven: Code will be triggered when event happens asynchronously

Use of framework



Node REPL: Node Read Evaluate Print Loop



To enter into Node REPL type "node" in terminal and can perform commands

.exit is to Exit or Ctrl+C (2 times)